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Studierichting: Informatica
Oefeningenreeks 5E nr. 2

$$S = 4\int_{\frac{\pi}{2}}^{0} |a\sin^3 t| \cdot |-3a\cos^2 t\sin t|\, dt$$

$$= -12a^2\int_{\frac{\pi}{2}}^{0} \sin^4 t\cos^2 t\, dt = 12a^2\int_0^{\frac{\pi}{2}} \sin^4 t\cos^2 t\, dt$$

Gebruik: $\sin^2 t = \dfrac{1-\cos 2t}{2}$ en $\cos^2 t = \dfrac{1+\cos 2t}{2}$

$$\Rightarrow \sin^4 t\cos^2 t = \left(\frac{1-\cos 2t}{2}\right)^2 \cdot \left(\frac{1+\cos 2t}{2}\right) = \frac{(1-\cos 2t)^2(1+\cos 2t)}{8}$$

$$S = 12a^2\cdot\frac{1}{8}\int_0^{\frac{\pi}{2}} (1-\cos 2t)^2(1+\cos 2t)\, dt = \frac{3a^2}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2t)^2(1+\cos 2t)\, dt$$

Bereken de integrand:

$$(1-\cos 2t)^2(1+\cos 2t) = (1-2\cos 2t+\cos^2 2t)(1+\cos 2t)$$

$$= 1-\cos 2t-\cos^2 2t+\cos^3 2t$$

$$S = \frac{3a^2}{2}\int_0^{\frac{\pi}{2}} \left(1-\cos 2t-\cos^2 2t+\cos^3 2t\right) dt$$

Gebruik bekende waarden:

$$\int_0^{\frac{\pi}{2}} \cos 2t\, dt = 0,\ \int_0^{\frac{\pi}{2}} \cos^2 2t\, dt = \frac{\pi}{4},\ \int_0^{\frac{\pi}{2}} \cos^3 2t\, dt = 0$$

$$S = \frac{3a^2}{2}\left(\frac{\pi}{2}-0-\frac{\pi}{4}+0\right) = \frac{3a^2}{2}\cdot\frac{\pi}{4} = \frac{3\pi a^2}{8}$$

References

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